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Optimization and characterization of single photon sources

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Abstract

Single photon sources play a key role in the development of tomorrow's quantum technologies, particularly in quantum computing and communication. The effectiveness of these sources depends on a number of criteria, of which efficiency is a key one. In the source developed by the company, the photonic crystal stands out as a key component with great potential for improving performance.

Inverse design, an approach based on mathematical methods, is now gaining in importance in the field of photonics. Unlike traditional design approaches, which start from specifications to create a device, inverse design works backward: it first defines the desired performance, and then uses optimization algorithms to find the best possible configuration of system parameters. This makes it possible to explore complex and counter-intuitive designs that are often inaccessible using conventional methods.

In my work, I applied this technique to optimize the transmission of the photonic crystal, using gradient descent via a method called the 'adjoint method'. My aim was to modulate certain parameters of the photonic crystal in order to maximise its transmission.

Key Words : Single Photon Source, Photonic Crystal, Inverse design, Optimization, Transmission

RÉSUMÉ

Les sources de photons uniques jouent un rôle essentiel dans l'essor des technologies quantiques de demain, notamment en informatique et en communication quantiques. L'efficacité de ces sources repose sur plusieurs critères, parmi lesquels le rendement occupe une place centrale. Dans la source développée par l'entreprise, le cristal photonique se distingue comme un composant clé avec un fort potentiel d'amélioration des performances.

L'inverse design, une approche basée sur des méthodes mathématiques, gagne aujourd'hui en importance dans le domaine de la photonique. Contrairement aux approches traditionnelles de conception, qui partent de spécifications pour créer un dispositif, l'inverse design fonctionne à rebours : il définit d'abord les performances souhaitées, puis utilise des algorithmes d'optimisation pour trouver la meilleure configuration possible des paramètres du système. Cela permet d'explorer des conceptions complexes et contre-intuitives, souvent inaccessibles par des méthodes classiques.

Dans le cadre de mon travail, j'ai appliqué cette technique pour optimiser la transmission du cristal photonique, en recourant à la descente de gradient via une méthode appelée "méthode de l'adjoint". Mon objectif était de moduler certains paramètres du cristal photonique afin d'en maximiser la transmission.

Mots Clés : Source de photons uniques, Cristal Photonique, Inverse design, Optimisation, Transmission

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First, I would like to thank Claus PEDERSEN for accepting my internship application for the company. Secondly, I am particularly grateful to Jeppe ROULUND for his mentorship and availability throughout my internship.

During this period, I appreciated working with you in this dynamic environment which is Sparrow. I have gained a lot of skills and knowledge which help me to improve my understanding of physics phenomena. Thanks to all of you for these three months.

Finally, I would like to thank my family for their help and support for this internship and in general for my studies.

NOMENCLATURE

FOM	Figure Of Merit
NW	Nanobeam Waveguide
PBG	Photonic Band Diagram
PhC	Photonic Crystal
PSO	Particle Swarm Optimization
PWE	Plane Wave Expansion
QD	Quantum Dot
SPS	Single Photon Source
TE	Transversal Electric
WG	Waveguide

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Introduction

This internship constitutes a component of my second-year education at Telecom Physique Strasbourg before my last year of master in quantum sciences and technologies in the University of Strasbourg. My internship has taken place at Sparrow Quantum, in Copenhagen (Denmark) a spin-off of the Niels Bohr Institute.

At the heart of my experience was the optimization of the transmission of a photonic crystal Waveguide, which is one of the key components of the single photon source that Sparrow Quantum produces.

My involvement in this project wasn't from its inception because some improvements had been already made before my internship. I followed the work of my supervisor Jeppe ROULUND and we worked very closely during my internship. However, the framework based on inverse design that I used during my internship was very new for the company. It was the first attempt for this kind of optimization.

CHAPTER 1

SPARROW QUANTUM

1.1 Presentation of Sparrow Quantum

Sparrow Quantum is a start-up company that focuses on providing deterministic single-photon sources. The company which is located in Copenhagen was founded in 2016 by Professor Peter Lodahl from the Niels Bohr Institute based on his work in the past 20 years. In the following years, the company continued to grow, and today there are about 20 employees.

Indeed, Sparrow Quantum is composed of different teams: a management team, a business team, a research and development team with numerous industry PhD and Postdocs, a characterization team, and an engineering team in which I take part during my internship. Also, Sparrow Quantum has a board of directors and consultants consisting of six people.

Figure 1.1: Picture with Sparrow Quantum's team

1.2 Technology

The technology of the company is the single-photon source (SPS) based on light-matter interaction using semi-conductor quantum dots (QD)[1].These emitters are coupled to a nanophotonic nanostructure composed of a photonic crystal waveguide (PhC WG)connected to a shallow etched grating-out coupler. This structure is fabricated in a GaAs membrane. The structure is shown in Figure 1.2

The technology principle of the single photon source is based on exciting the quantum dots. When the photons are emitted by the quantum dots, these photons are guided by the waveguide and they exit by the shallow etched grating outcoupler. The spontaneous emission of quantum dots which is necessary for SPS is enhanced by using the Purcell effect. Purcell effect consists of the enhancement of a quantum system's spontaneous emission rate by its environment. More specifically, the local density of photonic states at the emitter position is enhanced. In this technology, the Purcell effect is enhanced by the design of the PhC WG.

A single-photon emitter is characterized by various properties and requirements, including key photophysical aspects such as maximum count rate, lifetime, and linewidth at 4K. Additionally, for many applications, important requirements include photon purity, indistinguishability, and efficiency.[2]

These deterministic single-photon sources are necessary for many applications such as quantum communication and quantum computing. This technology can be used in quantum computing for drug discovery or quantum key distribution (QKD) in the field of quantum cryptography. Recently, the new partnership between Sparrow Quantum and ORCA, a quantum computing company, shows the relevance of this product in the quantum technology field.[3, 4]

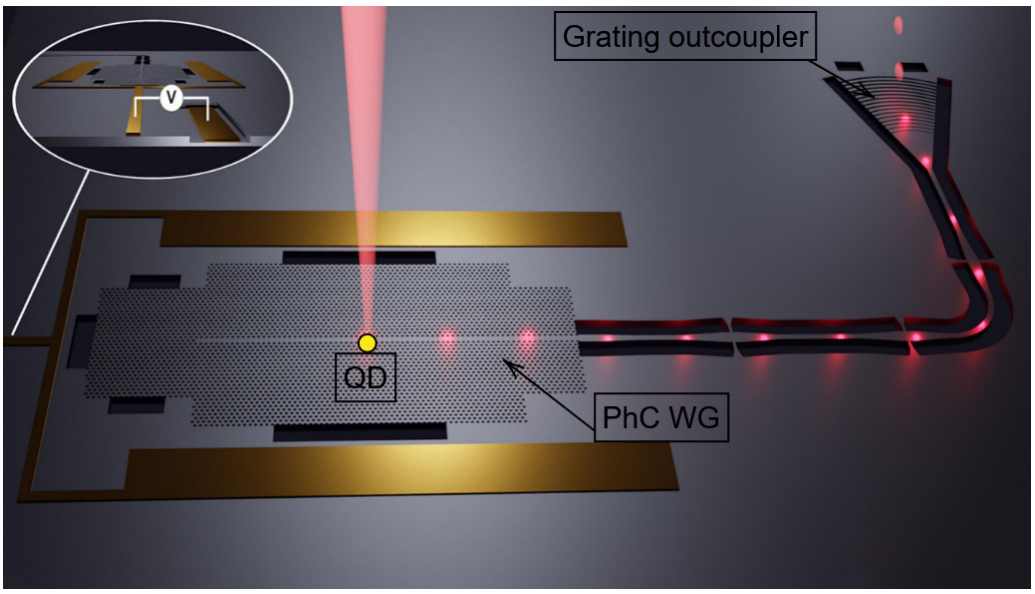


Figure 1.2: Overview of Sparrow Quantum's technology

CHAPTER 2

CONCEPTUAL FRAMEWORK

2.1 Explanation of the physical concept

As mentioned in the previous section 1.2, the single photon source is based on the excitation of a quantum dot. Quantum dots are nanoscale semiconductor particles which have unique optical and electronic properties. They are called "artificial atoms" and they can be simplified as a two-level system. A photon is produced during the recombination of an electron-hole pair.

Nowadays there are different ways to modify the photonic environment for deterministic generations of photons like micropillars, nanowires, bullseye cavities, or photonic crystals [5]. All of these experimental realizations need a strong fabrication to grow the quantum dot.

As we mentioned before, Sparrow Quantum uses a PhC defect waveguide to embed the quantum dot.

Quantum dots are fabricated using semiconductor materials with different bandgap energies, which are combined to form a heterostructure. The periodic potential experienced by electrons within the crystalline lattices of these materials leads to the formation of discrete energy levels within the bandgap, creating quantized energy states.

In this context, GaAs and InAs are commonly used to form such a heterostructure. Self-assembled quantum dots are grown by exploiting the intrinsic strain caused by the lattice mismatch between the two materials. This strain causes the material to form small dots due to the Stranski-Krastanov growth mode. The light blue region in Figure 2.1 represents the potential well created by the heterostructure. This potential well confines electrons in all three dimensions, resulting in discrete, quantized energy levels. These discrete energy levels are analogous to those in an atom, leading to quantum dot behavior.

To achieve the excitation of a QD, various methods can be used depending on the energy levels

of the QD. These methods include quasi-resonant excitation or p-shell excitation, and resonant excitation or s-shell excitation. Resonant excitation could be described as when a photon with the same energy as the bandgap of the QD is absorbed. A photon with the same energy is released after recombination. Non-resonant excitation involves discrete states of the QD; when excited, non-radiative relaxation occurs, leading as well to recombination.

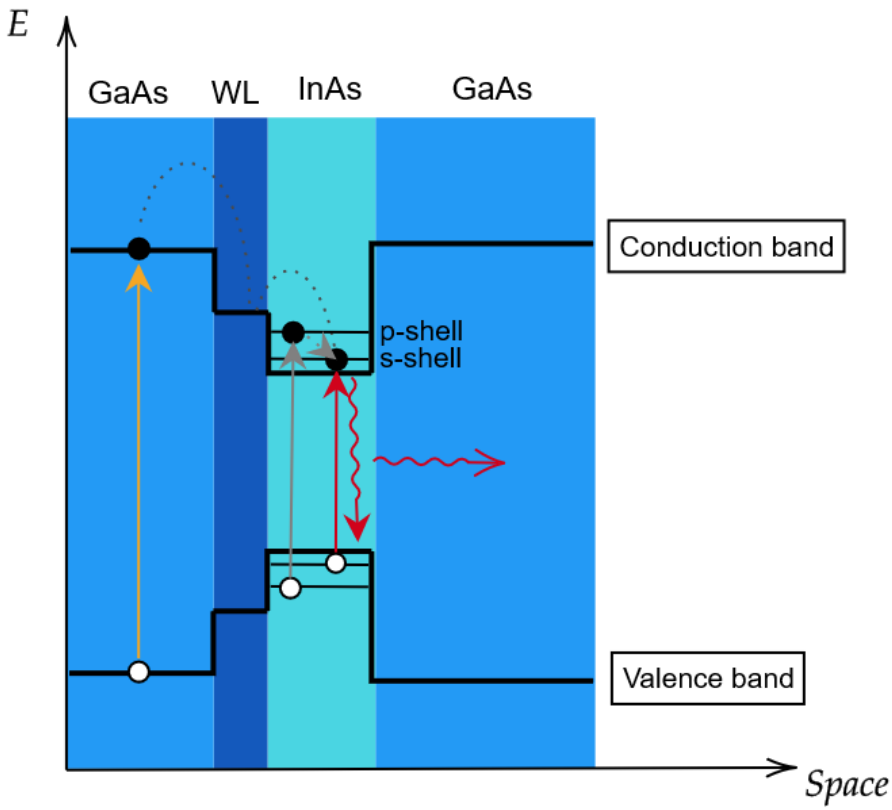


Figure 2.1: Diagram of the Quantum dot

In our case, when the photon is emitted, we want to couple the emission of the quantum dot into the waveguide mode and afterwards, it needs to be collected using a shallow etched grating coupler.

One of the most important features of the SPS is the efficiency. The efficiency of the quantum dot single photon source could be described as

$$\eta_{\text{SPS}} = \eta_{\text{QD}} \times \eta_{\text{Chip}} \times \eta_{\text{Coupling}} \tag{2.1}$$

Currently, some components of the current device need to be improved to improve this efficiency. My project during my internship was focused on the photonic crystal waveguide in which you can find the QD. The goal was to improve the transmission of the emitter in the fundamental mode TE, by doing an optimization of the geometry between the photonic crystal waveguide and the nanobeam waveguide.

First, let's define what a photonic crystal is. Photonic crystals (PhCs) were discovered in 1987 by S.John and E.Yablonovitch [6, 7]. Within PhCs, electromagnetic waves scatter. For some wavelengths and wave numbers, destructive interference occurs, resulting in the formation of a photonic bandgap (PBG), which is similar to the energy bandgap of electron waves in a semiconductor. PhCs are used in several notable applications such as optical sensing, photovoltaic applications.[8, 9]

One-dimensional (1D), two-dimensional (2D), and three-dimensional (3D) Phcs represent distinct dimensions of periodic structures designed to control the flow of light. In 1D PhCs, refractive index variations occur along a single direction, enabling precise control over light in that dimension. Two-dimensional PhCs expand this control to two dimensions, offering more complex light manipulation and enabling the creation of planar structures with applications in waveguides and microcavities.

The properties of this nanophotonic structure are described by its band diagram. Specifically, the photonic bandgap, a key feature of the band diagram controls what kind of light can exist within the crystal. A propagating waveguide mode can be introduced in the band diagram by removing a row of holes in the PhC, forming a PhC WG.

For PhC, some important parameters are refractive index, lattice constant (a) and hole radius(r).These parameters will change the band diagram.

The band structure of a GaAs/Air multilayer is in Figure 2.2. It looks like a photonic crystal in 1D.

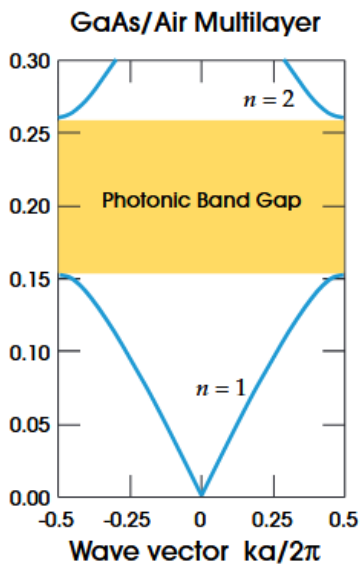


Figure 2.2: Band structure 1D from the book "Photonic Crystals: Molding the Flow of Light" [10]

For the device used on the chip, the band diagram is a band diagram of a photonic crystal slab which helps confine light in 3D by using a band gap in plane and an index confinement out of plane due to the internal reflection on the interface with the surrounding air or vacuum. In the fabrication process, the photonic crystal is fabricated after the growth of quantum dot in the semiconductor membrane and that is very convenient because QDs are highly sensitive to environmental condi-

tions during growth and if the PhC is fabricated before, it could introduce surface roughness and irregularities.

In figure 2.3 we have the configuration of a triangular lattice which is what we have in the component we simulate without the deleted row of holes.

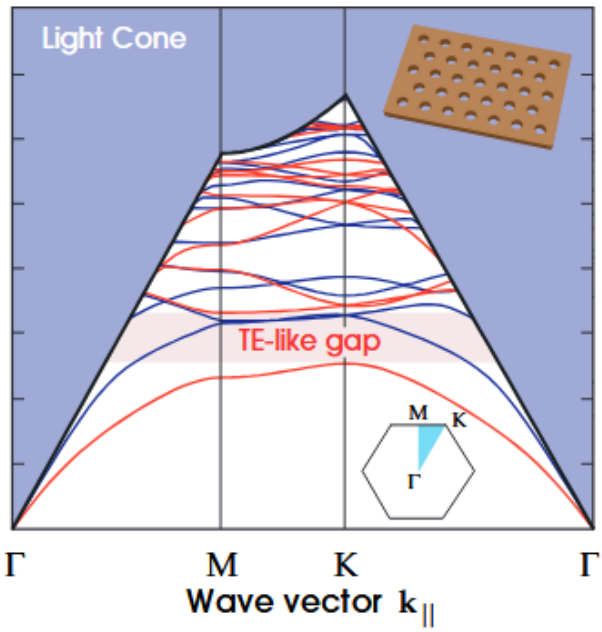


Figure 2.3: Band structure from the book "Photonic Crystals: Molding the Flow of Light" [10]

In figure 2.3, we can see a Transverse Electric mode bandgap, which means no light in the TE modes can exist in this region.

CHAPTER 3

WORK DURING MY INTERNSHIP

3.1 Context

My work was focused on the following components, the PhC WG and the NW.(nanobeam waveguide)

To begin, we want to optimize the transmission for a PhC WG with fixed parameters and we will see afterward if we can extrapolate the behavior for different a and r values.

Photonic crystal parameters with dispersive material:

- lattice constant $a = 246nm$
- holes radius $r = 77nm$
- hole material = etch, n=1
- membrane material=$GaAs$, n=3.465 at 940 nm
- inner membrane material=$Al0.3Ga0.7As$, n=3.275 at 940 nm

The photonic crystal I have to optimize has those parameters but we can use different parameters for the a or r values. It will change the key features of the PhC.

The geometry we simulate is not the geometry of the Block slow-fast section of PhC WG that we saw previously in figure1.2 because it is challenging to catch the transmission we want if we reproduce the device on the chip. Indeed, it is very hard to finish the geometry of the simulation if we simulate the Block slow-fast section. Thus to calculate the transmission we use a symmetric geometry which is easier to modelize.

Configuration of the geometry shown in figure 3.1:

- slow section consisting of 20 holes in x direction and 14 holes in y direction, with a lattice constant of $a_{slow} = a$

- two fast sections consisting of 11 holes in x and 10 holes in y with a lattice constant of $a_{fast} = 1.07 * a$. At wavelengths where the group velocity of the slow-light region is low, the fast-light waveguide supports a mode with a larger group velocity.
- two nanobeam waveguides; these are "white" rectangular are the etched parts that make the NW. They are in the same material as holes. That means if a hole is in this part they will overlap and don't affect the behavior

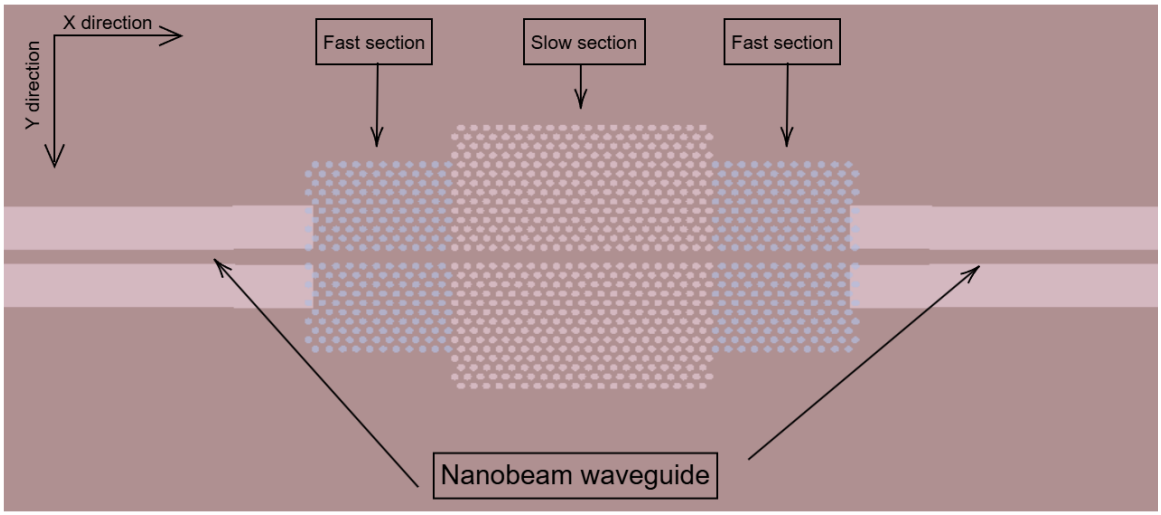


Figure 3.1: Geometrical configuration of the simulated component

In our case, the band diagram of the PhC WG is the following in figure 3.2.

The purple shaded area represents the slab modes of the bulk photonic crystal slab, and the gap between them is the bandgap.
Light which is confined in plane and out of plane is depicted as a light cone of radiation modes in yellow in the band diagram. At lower frequencies $\frac{ka}{2\pi} \rightarrow 0.5$ of the lower waveguide mode (k is the wave number), a dramatic change in the local density of states (LDOS) is observed, which abruptly drops to zero. This is referred to as the band edge of the photonic crystal. In PhCs, it refers to the spectral location at the boundary between allowed and forbidden modes of propagation for certain wavelengths. The waveguide modes are highly dispersive, with a group velocity approaching zero near the band edge, $V_g(\omega) = \frac{d\omega}{dk} \approx 0$, supporting slow light. The slowdown factor is determined by the group index $n_g(\omega) = \frac{c}{V_g}$, which is proportional to the Purcell factor[11]. That means in the slow section, the Purcell factor is enhanced due to the high group index.

The fast section has a larger group velocity. Indeed, by increasing the distance of the lattice constant with a uniform stretch factor. We can see that the group velocity of the slow-light region is low close to the band edge. At the same wavelength, the fast-light waveguide supports a mode with a larger group velocity as shown in figure 3.3 The transition between these two sections involves moving from a high group index to a low group index, which can create an abrupt interface. This interface may reduce transmission efficiency, so our goal is to optimize the transmission and

minimize any losses caused by this transition. The fast region takes as well the role of a mode adapter in the nanobeam waveguides.

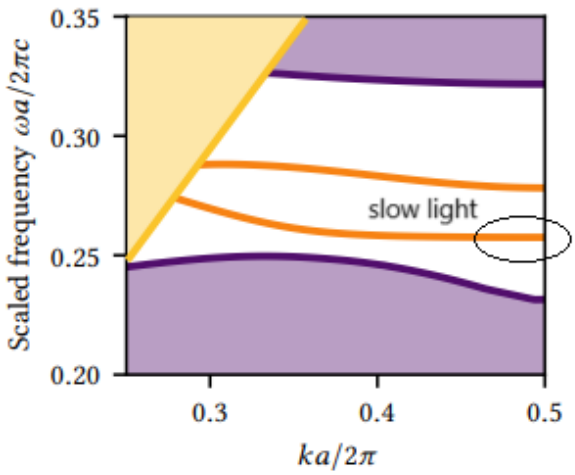


Figure 3.2: Band diagram of slab photonic crystal from [5] obtained by plane wave expansion method

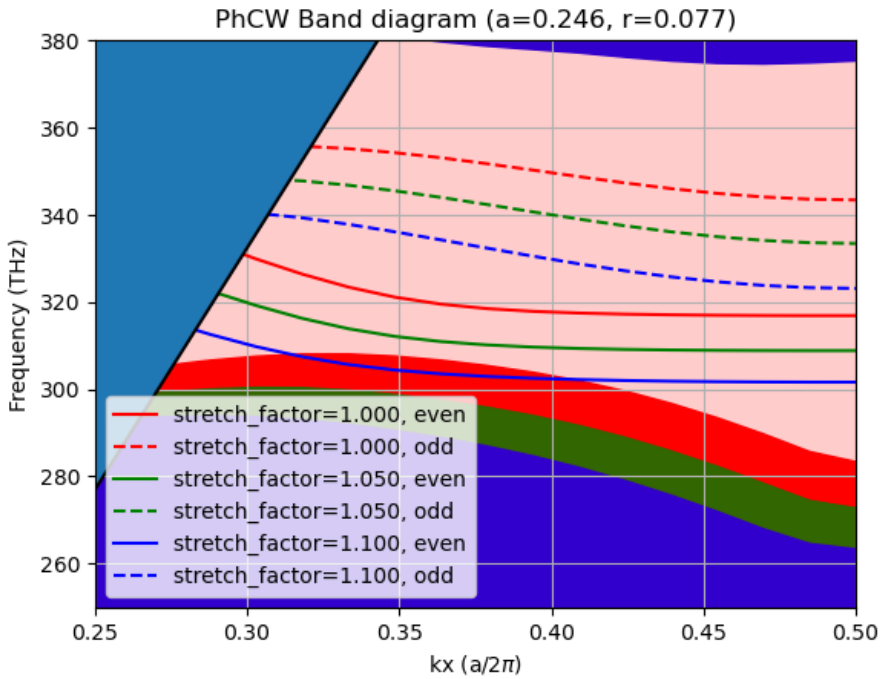


Figure 3.3: PHC WG band diagram

To improve this device, we need to optimize different parameters of the current geometry. Currently, the transmission on the current PhC WG is plotted in figure 3.4. This transmission figure is highly significant because it defines the objective we want to exceed. It serves as a critical reference point or baseline for evaluating better performance.

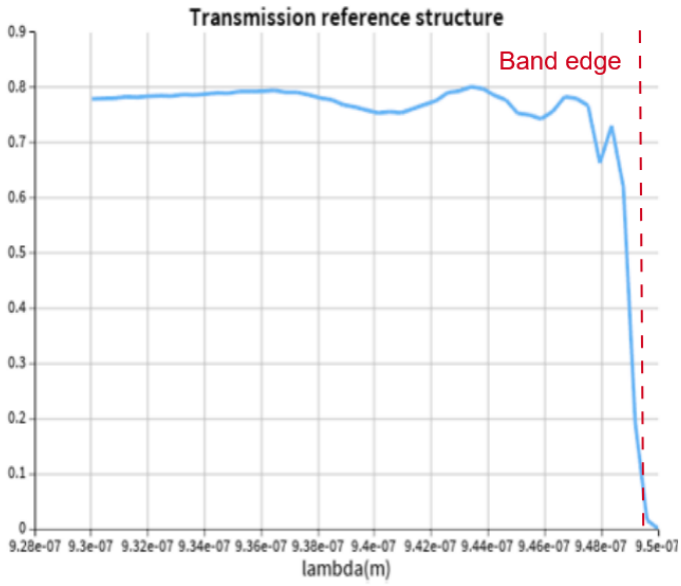


Figure 3.4: Transmission on reference structure

In section 2.1, we mentionned that the SPS efficiency is linked to chip efficiency.

$$\eta_{\text{Chip}} = \frac{1}{2}(1 + R_{\text{end}}) \times \eta_{\text{slow}}(L_{\text{slow}}) \times T_{\text{slow-fast}} \times \eta_{\text{fast}}(L_{\text{fast}}) \times T_{\text{fast-beam}} \times \eta_{\text{beam}}(L_{\text{beam}}) \times T_{\text{grating}} \quad (3.1)$$

in which there are differents terms :

- R_{end}: intensity reflection coefficient of the termination mirror of the waveguide.
- η_{slow}: intensity transmission of the slow-light section of the waveguide of length L_{slow}.
- $T_{\text{slow-fast}}$: intensity transmission of the slow-fast mode adapter section.
- η_{fast}: intensity transmission of the fast-light section of the waveguide of length L_{fast}.
- $T_{\text{fast-beam}}$: intensity transmission of the interface between photonic-crystal waveguide and nanobeam waveguide.
- η_{beam}: intensity transmission of the nanobeam waveguide of length L_{beam}.

My work was to improve the transmission $T_{\text{slow-fast}}$ and $T_{\text{fast-beam}}$. The behavior of these transmission are linked to the transmission plotted in figure3.22. The aim is to beat our baseline to improve η_{Chip}.

3.2 Simulation Framework

My internship aims to improve the transmission between a PhC WG and an NW. Also, to simulate the behavior of the interface between the PHC WG and the NW we used the Ansys Lumerical software in which we used the Finite-Difference Time-Domain (FDTD) method. FDTD is a numerical analysis technique used for modeling computational electrodynamics. It works by solving Maxwell's equations on a discrete grid over time, allowing for the simulation of how electromagnetic

waves propagate through different materials and structures. It works by dividing space and time into a grid of discrete points and approximating the derivatives in Maxwell's equations using finite differences. The method employs a scheme, where the electric (E) fields at a given time t are computed based on the E fields at the previous time step t - Δt and the magnetic (H) fields at t - $\Delta t/2$ This process is alternated to update both the E and H fields, allowing the field patterns to be propagated through time, offset by half a time step Δt.

In the previous section 3.1, we define the main part of our simulation. For the several simulations, we used a mesh accuracy of 10 points per wavelength, which allows us to understand the behavior of change in parameters.

MEMBRANE

We can see in figure 3.5, etched holes because we can see them as pillars and the different membranes of the component.

Figure 3.5: Profile View of PhC WG

MODE

We want to improve the transmission through the photonic crystal into the fundamental TE mode of the output waveguide. The TE mode profile can be seen in figure 3.6.

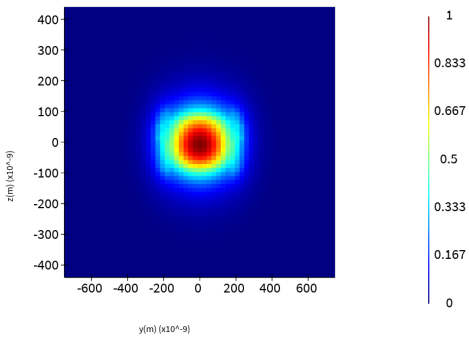
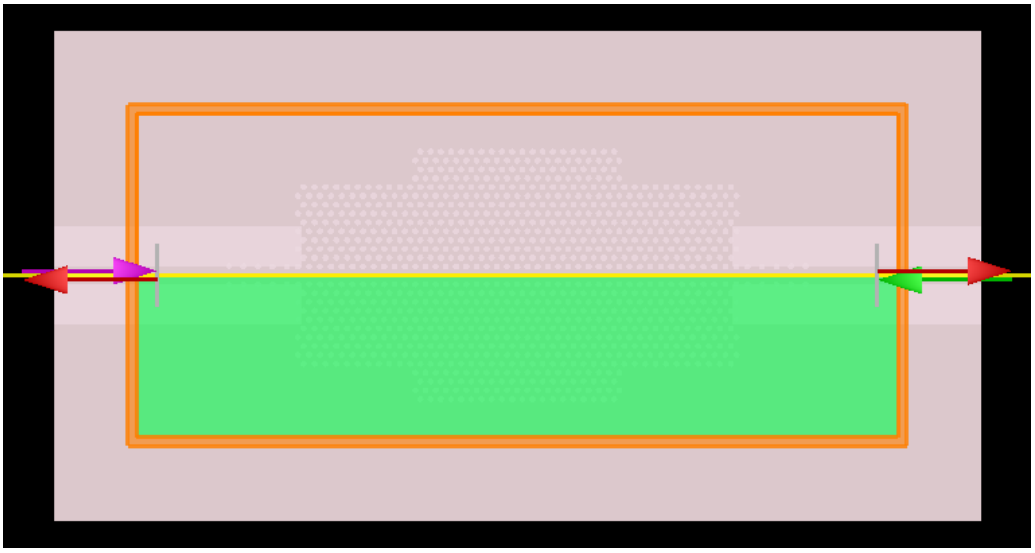


Figure 3.6: Fundamental mode TE

To run a simulation, we also need to define the region of the FDTD with some conditions, a source, and different monitors in which we can get results like the transmission. All of these components are necessary for a simulation , this is shown in figure 3.7

Figure 3.7: Different components for the simulation



Orange: FDTD region Left Side: Source Right Side: Monitor

3.3 Improvements done

Different ways to improve the transmission have been already done but we want to try an inverse optimization to improve relevant parameters.

In the literature, there are several ways to improve the coupling between a PhC WG and a NW. Some common improvements are tapered radius holes of the PhC WG, tapered distance holes of the PhC WG, and changes in the termination at the interface.[12, 13]. Some of these possibilities are not very relevant for us because fabricated hole radii have significant uncertainty. Therefore tinkering with multiple sizes of holes would not be very robust.

In addition, we want to improve the transmission in a specific mode of light so we need to be aware of the band diagram of the PhC because if we change the geometry of the fast section the band diagram is changed because guided modes are not distinct, it can affect the transmission.

Improvement of PhC WG and NW termination

This improvement plays a major role in improving the transmission[13] Indeed, this kind of improvement plays with the position of the first column of holes and NW. It can increase the transmission.

Here for the same geometry but a different position of NW, we have a different transmission. The "phase" means that positions of holes are set up regarding the kind of rows, either odd or even rows. The number of rows begins at 1.
Figures **??** and **??** depict two distinct configurations of the phase.

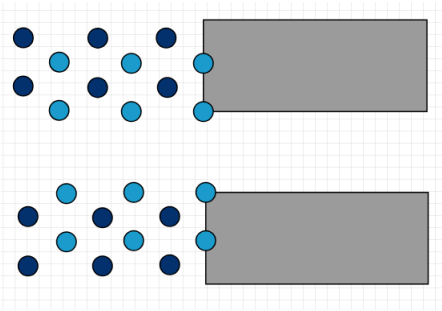


Figure 3.8: Geometry odd phase

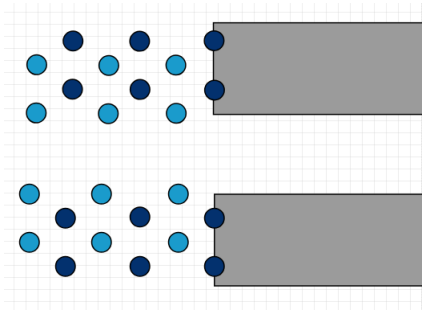


Figure 3.9: Geometry even phase

In figures 3.10 and 3.11, we plot the transmission for the both position and we can see that the even phase is better than the odd phase.

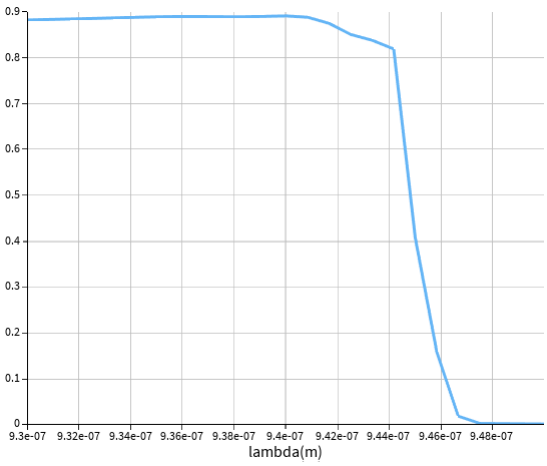


Figure 3.10: Transmission odd phase

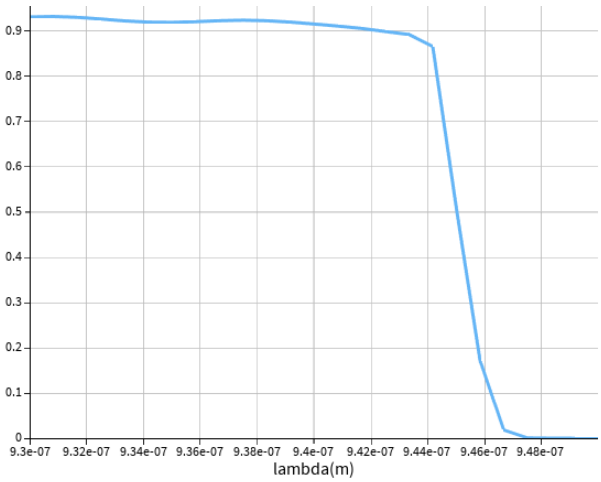


Figure 3.11: Transmission even phase

After the understanding of the relevance of the NW connection, a particle swarn optimization (PSO) had been tried regarding the position of holes and this PSO improved the transmission. PSO is a population-based stochastic optimization technique with the goal is to minimizing a function. These particles are subject to their behavior, the group behavior, and another parameter called an inertial weight, proportional to the velocity. All of these parameters need to be optimized according to the others to improve all behavior. In next section on the figure 3.13 ,we can see the position of holes which is optimized.

"Tapered stretch factor"

By tapering the stretch factor of holes in the fast section it leads to an improvement in the transmission. Indeed, without any taper between the slow section and fast section, there is an abrupt interface that affects the coupling efficiency.[14]
Thus, by doing an adiabatic transition at the interface between the slow section and the fast section it improves the transmission. Figure 3.13 depicts the geometry of these stretch factors. Firstly, the stretch factor is set around the same as in the slow section and this stretch factor is increased along

the fast section following a quadratic curve such as it is depicted in figure 3.12. Thus, we don't have an abrupt interface between both sections.

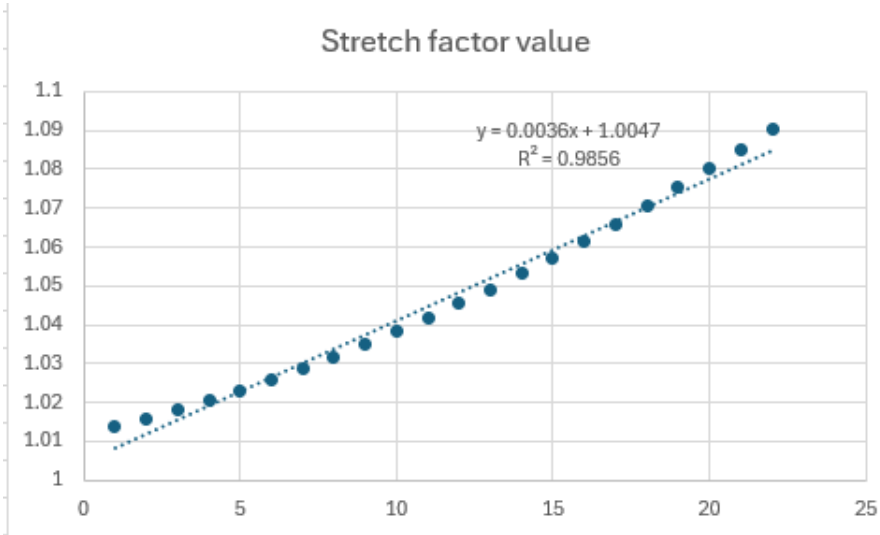


Figure 3.12: Polynomial stretch factor for different positions

Depending on the index numbers, the stretch factor is for even or odd columns. It is necessary in order to have a shift of holes and maintain the quadratic curve.

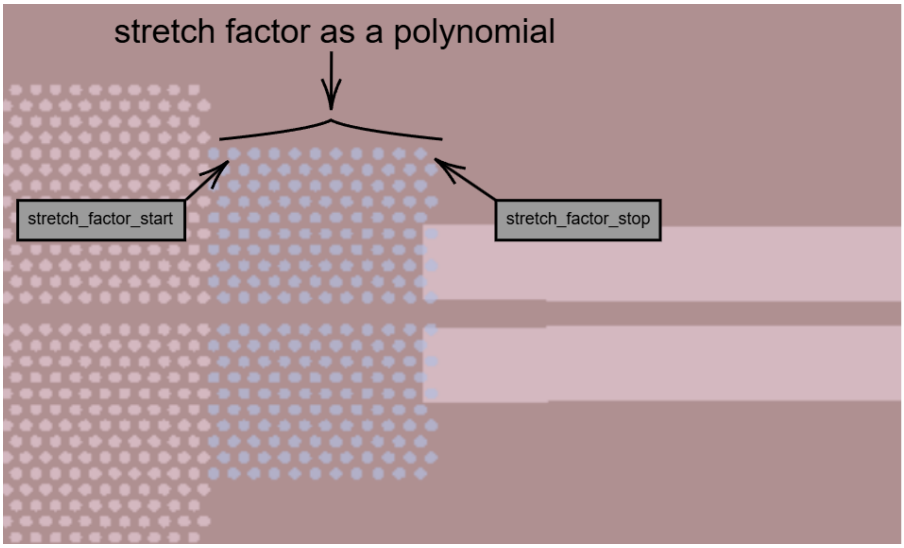


Figure 3.13: Quadratic stretch factor

Currently, the band diagram with the stretch factor tapered in the fast section is shown in the figure 3.3 We can see that guided modes odd and even are distinct, which is very interesting for the environnment of QD. Indeed, if modes overlap it can affect the field of the QD and the coupling efficiency between waveguide mode and emisssion of the QD. With this kind of improvement, the transmission reached is shown in figure **??**.This is a combination of two improvements.

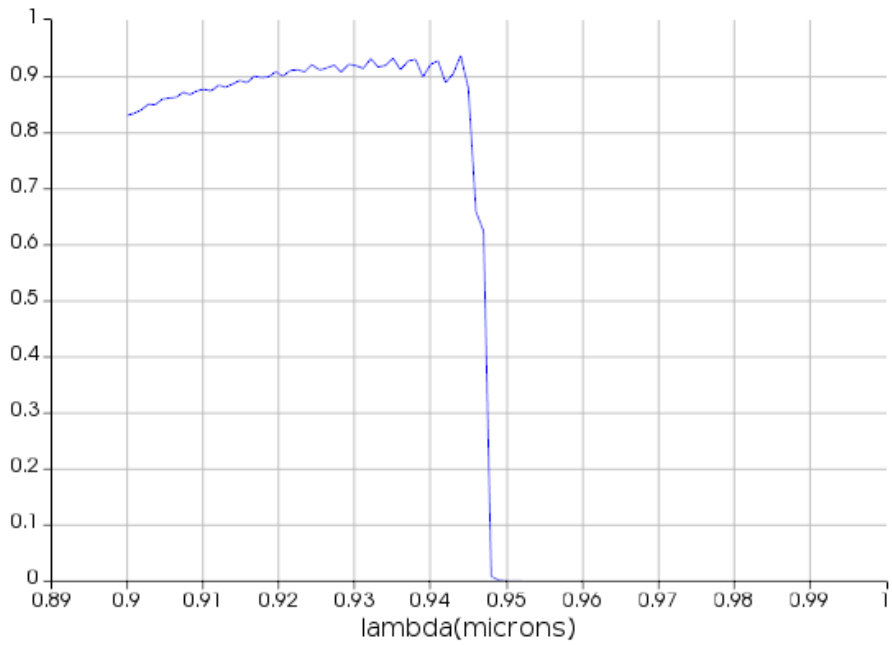


Figure 3.14: Transmission PhC WG quadratic stretch factor and NW termination

Tapered PhC

Another improvement could be to taper the position of holes in the Y direction because some papers demonstrate that could improve transmission.[12]

In our case, this improvement has been investigated but for some reason, we did not choose this possibility regarding the band diagram of this structure.

3.4 First Attempt: adding nanobeam holes

By using Lumerical, we want to improve the transmission with a first try in which we put some holes in the NW based on the attempt "Tapered Stretch factor". It looks like we want to increase the length of the PhC to improve transmission. Indeed, we assume that extending the PhC WG to an infinite length could significantly enhance transmission.

Set up

To set this line of holes, we describe this as a straight line like $y = c_{\text{ang}} * (x - x_0) + y_0$. In our tests, we can move different parameters of this line of holes: angular factor (c_{ang}), periodicity between holes, and the offset (x_0). The offset refers to the beginning position of the first hole in the straight line. Figure 3.15 is a figure of the try.

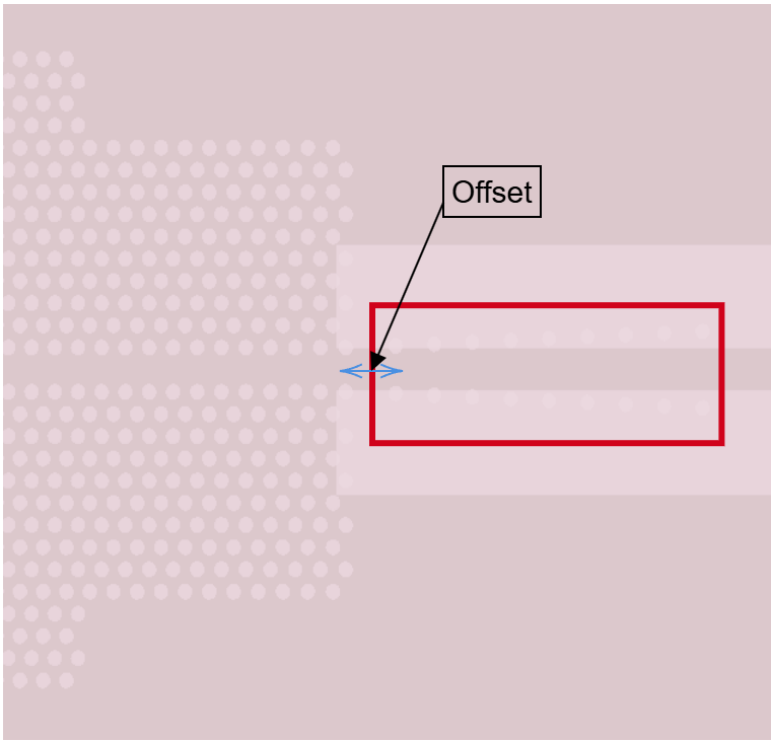


Figure 3.15: Geometry with nanobeam holes

Results

Firstly, we try to sweep the (c_{ang}) value of the line to understand effects on the transmission. However, a big increase in the angular won't change it a lot as it is shown in 3.16. Indeed, the etched parts and holes are composed of the same material, so if they are in the same space they will be overlapped.

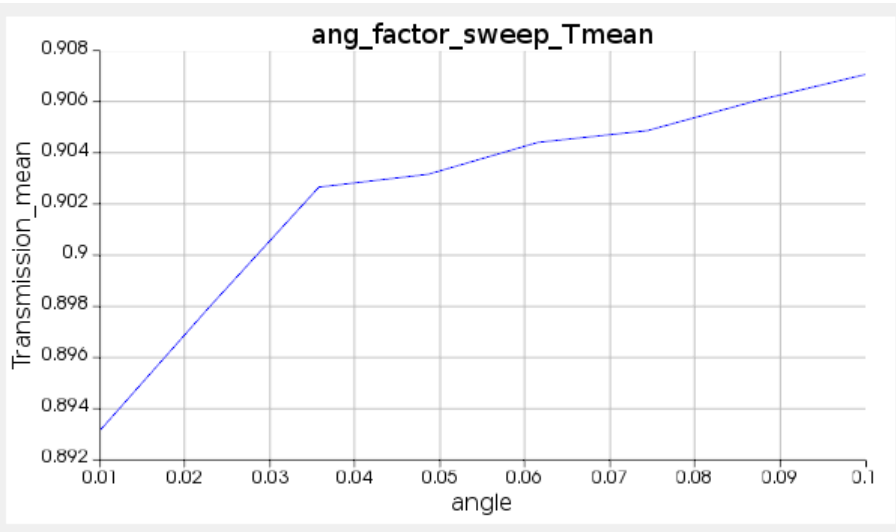


Figure 3.16: Sweep of the angular

The angular factor looks to increase a bit the transmission but we cannot increase more this factor because holes and etched parts will overlap. After the first results and the trend of them, we try an optimization by using Lumerical. Indeed, Lumerical has different methods to optimize a function. Here, we want to improve transmission by using particle swarm optimization. In the figure 3.17, the results show the best value of FOM that we obtain with PSO.

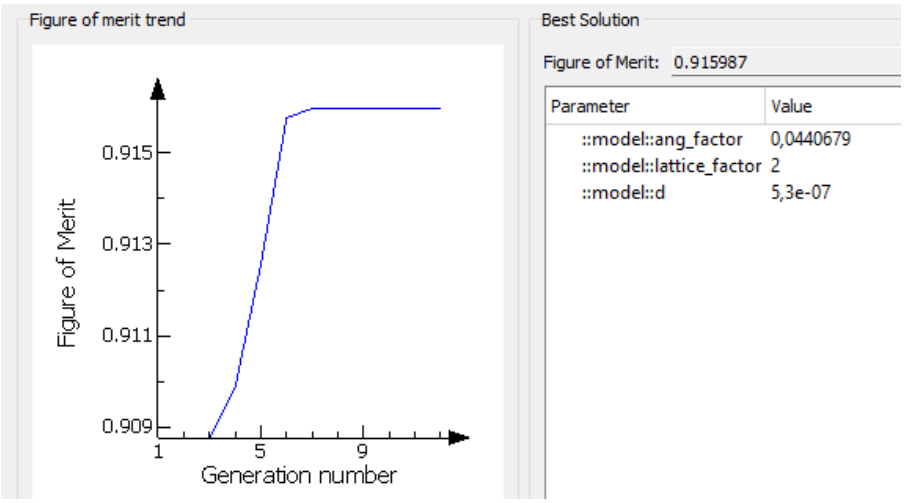


Figure 3.17: PSO optimization on try based on nanobeam holes

The *lattice factor* is the offset based on the periodicity of holes which is d, in the figure. However, it is not a big increase compare to the quadratic stretch factor geometry. We have the same trend of figure of merit with the PSO algorithm. That means, it doesn't change a lot when you add these nanobeam holes, except the angular factor.

3.5 Attempts based on literature

In the literature, there are different ways to improve coupling between WG and PhC WG as we mentioned before in section 3.3

According to M. Askari and A. Adibi [15], they show that when you taper the hole radius of your PhC WG, you can improve your coupling. Or you can minimize the radius of the first column of circles at the interface to improve coupling.

Results

Moving the size of the radius hole

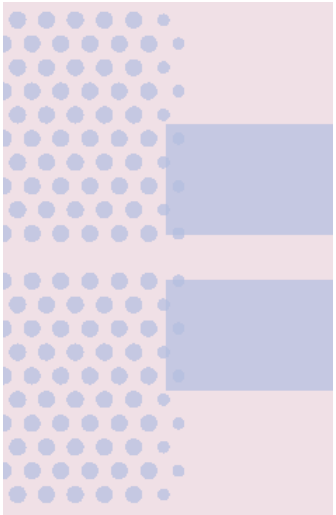


Figure 3.18: Changing radius size

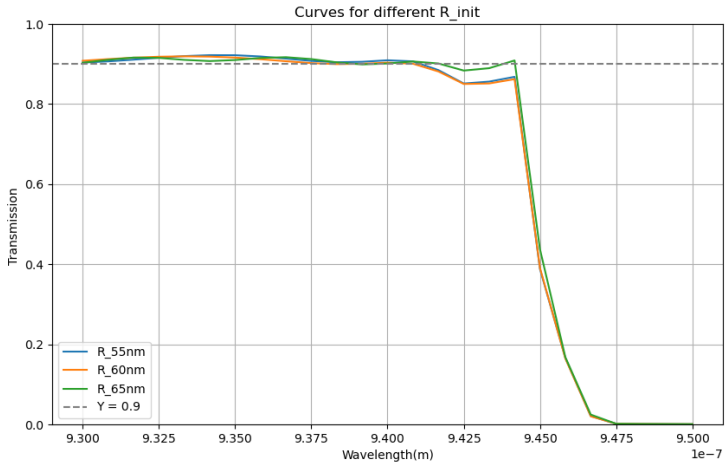


Figure 3.19: Curves of different initial radii

In figure 3.18, we can see that we move the initial radius and the transmission for diffrent radius is plotted in figure 3.19. The blue line considers a 55 nm radius which is better in the range of 930 until 940 nm but the green one is better close to the band edge.It is a radius of 65nm. As mentioned before, due to uncertainties during fabrication, we would prefer not to have multiple hole radii in the same nanostructure.

Changing the interface geometry

We will try to change the shape of the PHC WG and the etch part of the nanobeam waveguide to know if it improves transmission. Figure 3.20 shows the interface which is suggested in [15].

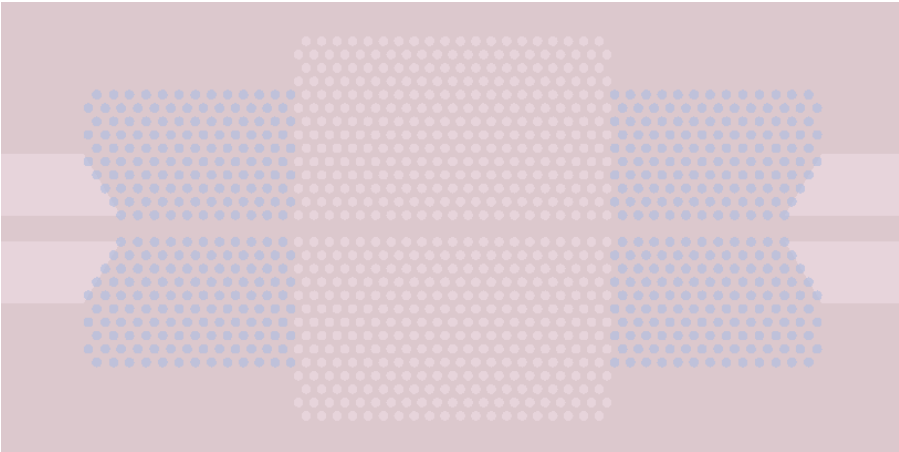


Figure 3.20: Changing the interface geometry between the NW and the fast region

By adjusting the interface of the photonic crystal and modifying the aperture at the interface, one could expect an improvement in transmission. It is similar to what we have done with nanobeam holes, but not on the NW. We assume that by adjusting the angle at the interface, light confinement can be enhanced.
Firstly, we check the transmission with 2 definitions of geometry. It is plotted in figure 3.21

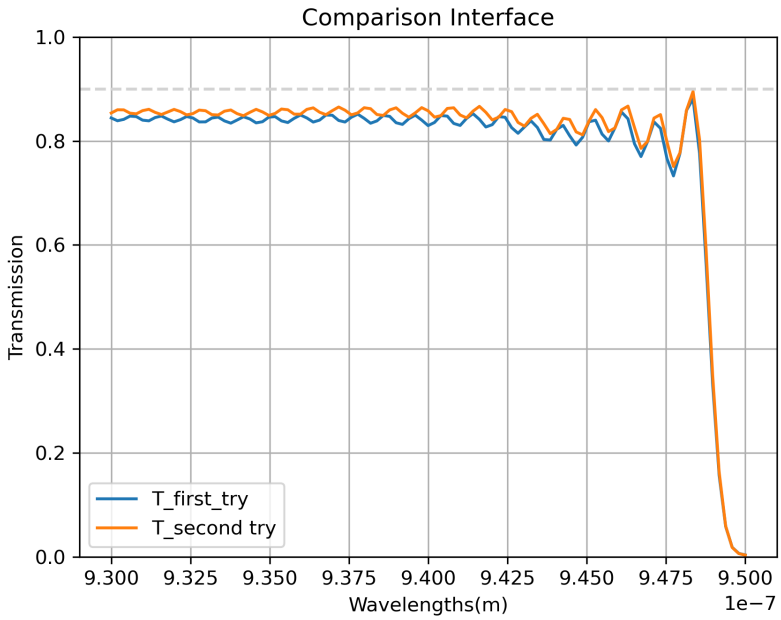


Figure 3.21: Transmission for two different definitions based on shape from paper from M. Askari and A. Adibi

Results are not very good for a first check compared to "Tapered Stretch factor" in figure **??** so we will focus on the inverse design. We could try a PSO optimization on angle and NW termination but the first check was not so good. Thus, it isn't worth it. Moreover, by playing on the angle or on the number of holes on the Y direction it could change the band diagram.

3.6 Inverse design of photonic device

Currently, photonic devices are becoming more and more complex with a higher integration in a lot of systems.[16] For optimizing photonic devices, we can use an inverse design to improve features of our devices. The concept of inverse design reverses the conventional design process, allowing new materials and compounds to be “reverse-engineered” simply by inputting a set of desired properties and characteristics and then using an optimization algorithm to generate a design that satisfies the desired properties.

In order to optimize parameters, we use the lumopt module. Lumopt requires lumapi which connects Python and Lumerical. Thus we have to reproduce the geometry done in the last try but with the Python API.
Indeed, the lumopt module uses Lumerical FDTD as a client to solve Maxwell’s equations and scipy libraries to perform the optimization. There are a lot of examples in this repository [17] that are very useful for beginners. We use a mathematical method based on gradient descent optimization. The calculation of the gradient is based on the adjoint method which is very efficient and it requires only 2 FDTD simulations. [18]

We base our optimization on what had been done before my internship. It will be our initial guess because the gradient descent optimization is very sensitive to the starting point.

In figure 3.22, this is the same transmission as the reference figure 3.13, but with 100 points and plotted between 930 and 950 nm.

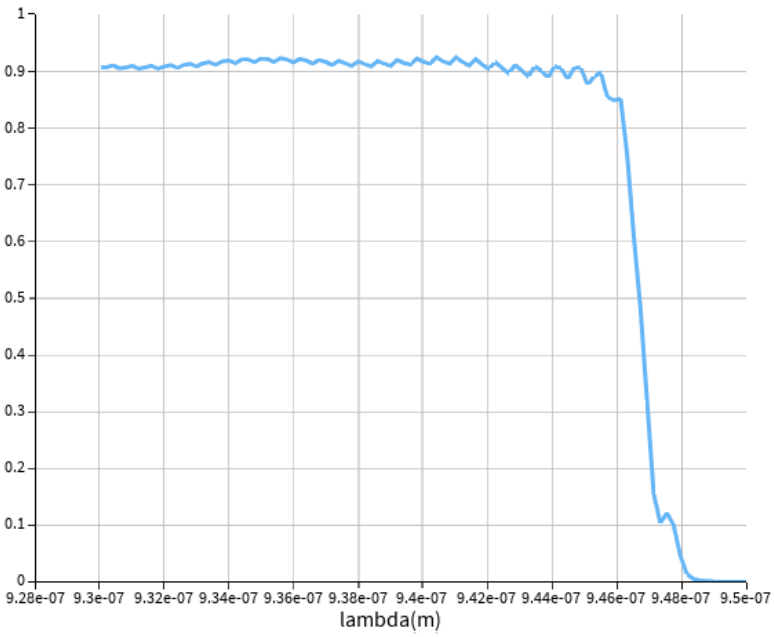


Figure 3.22: Transmission with Quadratic Stretch Factor with an increased resolution focused between 930–950 nm

Process of the Optimization

Here, we will show an example of how inverse design works. In inverse design, there are two manners of improving a cost function either by doing an inverse design based on shape/parameterized or a topology optimization. The shape optimization is more constrained than the topology optimization in which the refractive index of each discrete point in a defined region can freely vary between two values, it will change the shape regarding the topology. It leads to very fancy shapes and sometimes this is not manufacturable. With shape optimization key parameters defining the geometry of the object are varied systematically within defined bounds to identify the best-performing shape. Figure 3.23 shows an overview of the main difference between shape and topology optimization.

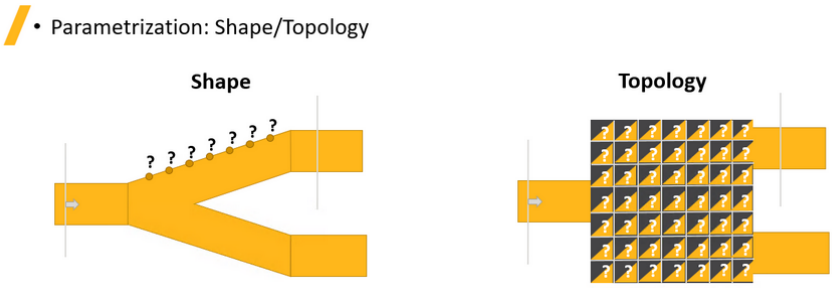


Figure 3.23: Inverse design based on shape and topology optimization; figure from Ansys website [19]

In our case, we want to maximize but it is the same process. Our configuration aims to optimize the FOM, which is the transmission in the PhC.

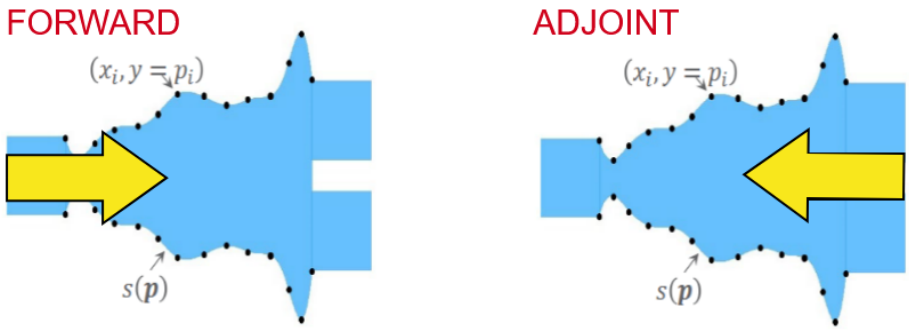


Figure 3.24: Two simulations, on the left it is the forward problem and in the right it is the adjoint problem which are necessary for optimization; figure from Ansys website [19]

The adjoint method is remarkable because it is able to optimize an exceedingly large number of parameters in a relatively small number of simulations. It is only two simulations for one iteration.It will be defined like this kind of process.

Forward Problem: The process begins with solving the forward electromagnetic problem using Maxwell's equations. This involves calculating the electromagnetic fields in the photonic structure

for a given design. The goal is to understand how light interacts with the structure based on its current design.

Adjoint Problem: After solving the forward problem, the adjoint method involves setting up and solving an adjoint problem. This problem is derived from the same Maxwell's equations but is tailored to provide sensitive information about the system. Specifically, it calculates how small changes in the material properties or geometry of the structure affect a defined objective function, such as transmission efficiency.

Gradient Calculation: The solution to the adjoint problem is then used to efficiently compute the gradient of the objective function with respect to the design variables, such as the refractive index distribution or the shape of the photonic structure. This gradient indicates how to modify the design to improve performance. In our case, we use an improved method of the gradient descent which is known as L-BFGS-B method.[20] This "quasi Newton method" uses an approximation of the Hessian matrix which is a square matrix of second-order partial derivative to find minima of a cost function. This optimization method uses this gradient information to update the design variables iteratively. Specifically, in each iteration, the design variables are adjusted by moving in the direction opposite to the gradient (the direction of steepest descent) by a small step size. In our case

$$\mathbf{x}_{\text{new}} = \mathbf{x}_{\text{old}} - \alpha \nabla f(\mathbf{x}_{\text{old}}) \tag{3.2}$$

- $\mathbf{x}_{\text{new}}$: The updated design variables after one iteration of gradient descent.
- $\mathbf{x}_{\text{old}}$: The current design variables before the update.
- α: The learning rate, which controls the step size in the direction of the gradient.
- $\nabla f(\mathbf{x}_{\text{old}})$: The gradient of the objective function with respect to the design variables at the current step. It indicates the direction of the steepest ascent, so subtracting it moves towards the steepest descent. In our case, it is the approximation of the Hessian matrix which is used to calculate the gradient.

Inverse design based on shape optimization

In this section, we will focus on an inverse design based on shape optimization by considering the last try before my internship.

Idea of Shape Optimization

Shape optimization or Parameterized optimization is a field of study that focuses on finding the best geometric configuration of an object or structure to optimize a specific performance criterion. With shape optimization we directly define the geometry parameters to vary, and the bounds of these parameters. It is less free than topology optimization.

Key Concepts:

Objective Function: This is the performance metric that needs to be optimized, maximized in our case, such as transmission.

Design Parameters: These are the parameters that define the shape of the object and can be adjusted during the optimization process, such as the dimensions between holes of the PhC which are linked to quadratic stretch factors of our "initial guess".

Constraints: These are limitations or requirements the optimized shape must satisfy, like our boundaries. These constraints are based on our "initial guess"

Optimization Algorithm: Mathematical methods or computational algorithms are used to iteratively adjust the design variables to find the optimal shape that meets the objective function while satisfying the constraint. In our case, we use the L-BFGS-B method with the adjoint method that we explained before.

Process of my optimization

We base our optimization on the last try done before my internship named as "Tapered stretch factor". It is one of the improvement done in a previous section 3.3. It will be our "initial guess" because the gradient descent optimization is very sensitive to initial parameters. This kind of optimization is very good for finding local minima but it can be stuck in this place. The last try consists of positioning the holes of the fast regions with an increasing gradient array of stretch factors following a quadratic curve. This stretch factor will be multiplicated by the lattice constant a. That leads to an increase in the distance between two holes in a fast region. This kind of geometry improves a lot the transmission of the PhC WG, as we saw previously in figure 3.22.

For our optimization, we define the stretch factor as parameters of our cost function. It is depicted in figure 3.25. The process of our optimization is shown in figure 3.26

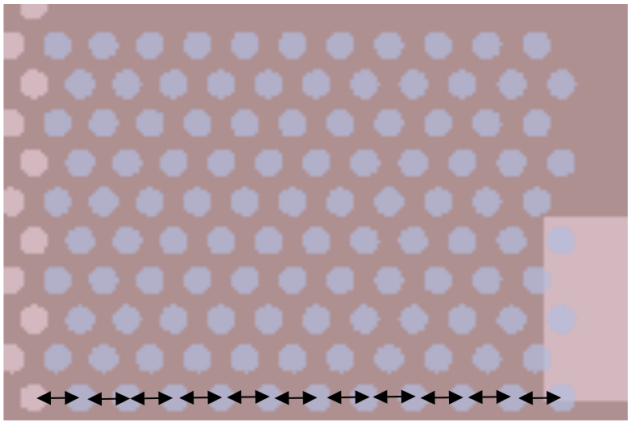


Figure 3.25: Stretch factors as parameters

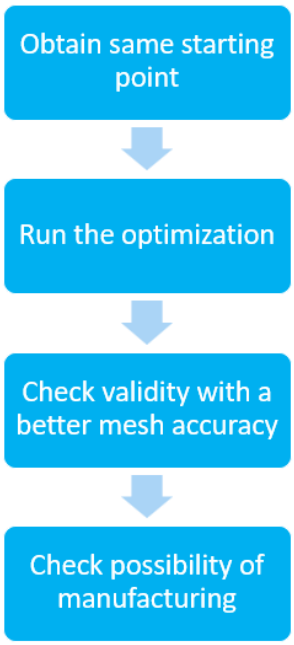


Figure 3.26: Optimization workflow

For the transmission curves in the optimization workflow, we define our simulation wavelength to 100 points between 930 and 950 nm.The amount of points is necessary to be the same to compare each curve.

Same Starting Point

Firstly, before running an optimization, we need to check that we obtain the same starting point with Lumerical and Python by using the lumapi, it would mean that we have the same geometry. If so, we can begin our optimization with a good initial guess. Moreover, we cannot use the same components in Python API for setting up because our optimization code is based on predefined components which are not the same as in the Lumerical file. To check the transmission in the mode we want, we need to use a mode expansion monitor because we want only one mode. In the Lumerical file, to create the source and to get the result of transmission they used ports. We can notice that the way we define our components affects a lot the FOM so we need to be consistent and take a reference that is defined with similar components otherwise we cannot compare.

In the figure below, there are two ways of defining the geometry of the fast section.
On one hand, using a geometry in which holes are defined by a stretch factor is the same way it was defined before my internship in which we follow polynomial values of stretch factor. We will name it "Geometry A".
On the other hand, holes are defined by using a stretch factor based in which rows they are placed. If they belong to odd rows, holes are defined as a middle value of the other holes of the even rows. We will name it "Geometry B".The construction of the geometry is shown in figure 3.27.
We do not notice a huge difference between the shapes but some parameters are shifted a little bit.

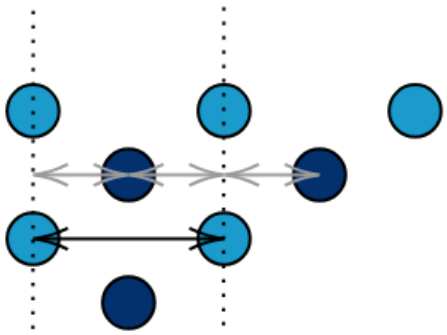


Figure 3.27: Construction of Geometry B

With geometries A and B, we obtain the same results in which curves overlap as it is evident in figure 3.28; that means we have the good initial guess and we can begin the optimization.

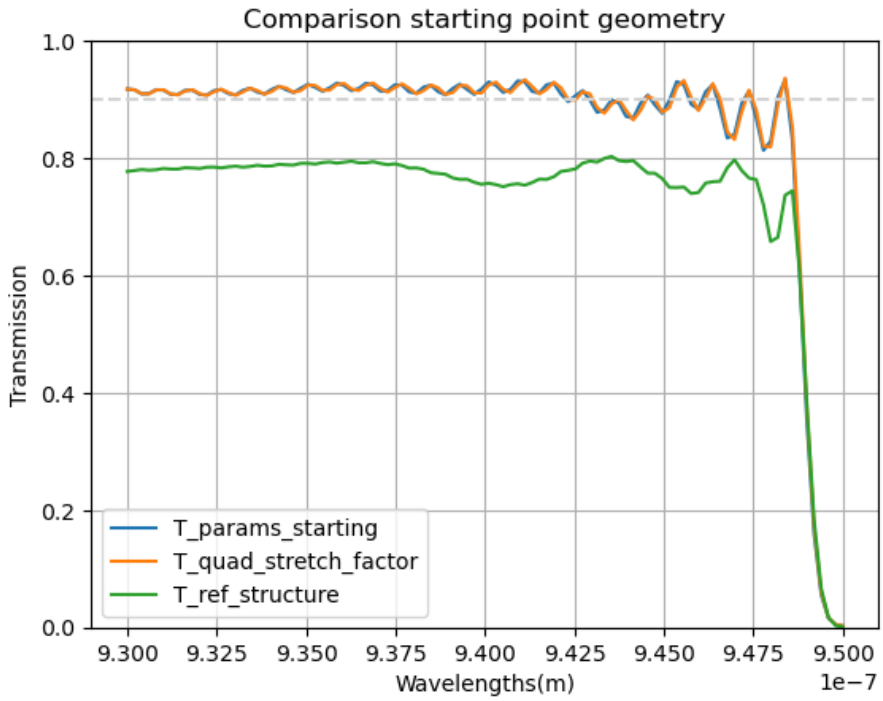


Figure 3.28: Starting point

We notice in figure 3.28 that there are a lot of oscillations. It depends a lot on the mesh size that we choose on the monitor named "FOM". It is not physical that the transmission oscillates but for the optimization, we will fix a mesh size. We can see here that we get the same transmission for Lumerical and Python. Considering this, we can use our optimization code on file as our initial guess. These spectra will be our reference and we have to beat those to improve transmission.

Optimization

In the optimization code, you first set up all necessary components and create a function that initially defines your geometry. This is what we obtained in the previous subsection 3.6.After the first iteration, this function will update the geometry based on the optimization process in which the Cost Function wants to be improved. It moves the position of holes based the stretch factor in order to improve the FOM. The optimization code is in appendix A.

Figure 3.29 shows what Python plots during our optimization.

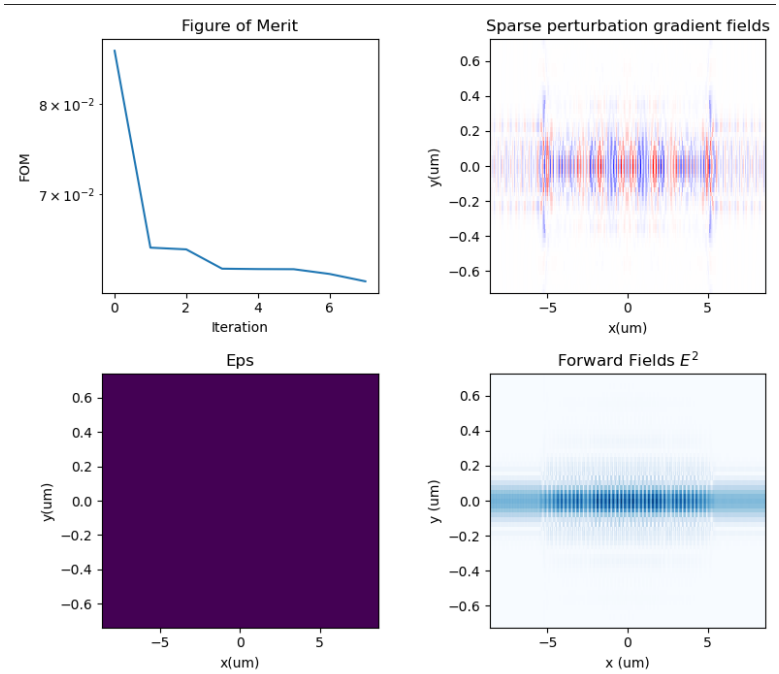


Figure 3.29: Plots of Python during optimization

For this first optimization, we use a mesh accuracy of 10 points per wavelength which allows us to understand the behavior of our optimization. However, when we figure out which definition of our geometry is the best regarding the optimization, we will try another optimization with a better mesh accuracy. It will be around 18 points per wavelength.

Optimization: using Geometry A and B

Here, we compare the optimization for both geometries.

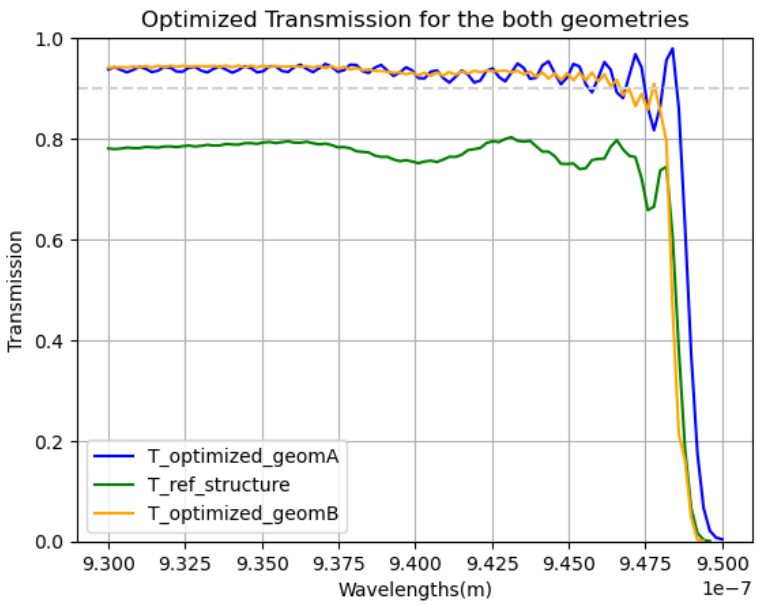


Figure 3.30: Plot of optimized structure for both geometries

In figure 3.30, we notice that the transmission with the geometry A is not really stable close to the band edge.

In the case of geometry B, the transmission is more stable close to the band edge.There are not a lot of oscillations compare to what we have with geometry A. Thus, we use geometry B for the rest of the checks. Moreover, it will be easier for defining the mask because it is the same way that the software defines it.

Better mesh accuracy

In order, to understand the behavior in a better manner we need to figure out if the behavior remains the same for an upper mesh accuracy. Figure 3.31 shows the comparison. Otherwise, it could be just an improvement for only 10 points per wavelength.

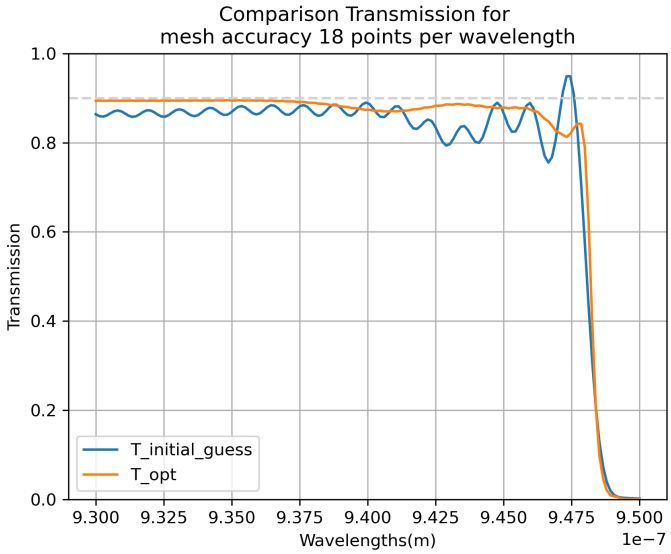


Figure 3.31: Comparison with a upper mesh accuracy

The behavior of the optimization is still the same for better mesh accuracy. Thus we can check for the fabrication requirements.

Fabrication requirements

At Sparrow, the positional uncertainty of the lithography step during fabrication is 3 nm. To calculate the transmission we round the stretch factor to three significant digits because 0,01*a =2,46 nm and it is the minimum accuracy for the fabrication. Thus we have just rounded the parameters in order to figure out if it is the same behavior with a truncation. This is illustrated in the figure 3.32.

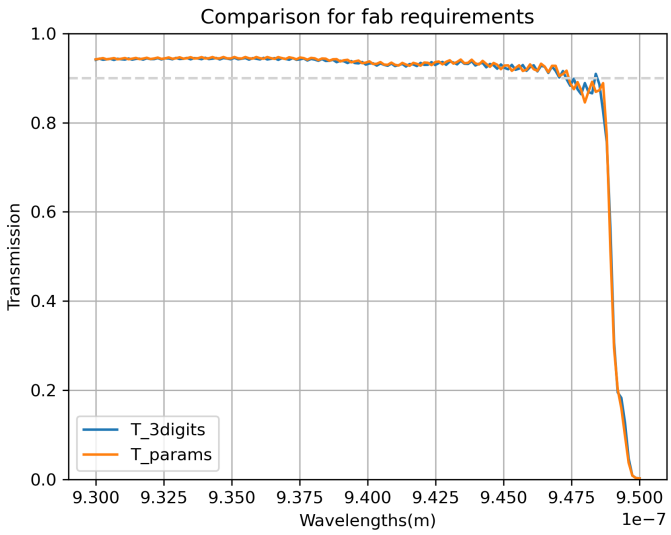


Figure 3.32: Behavior with fabrication requirements

To improve our optimization, an improvement could be to reduce the range of wavelength values.Indeed, by narrowing the bandwidth, we can enhance the FOM because the range between 945 and 950 nm could play a crucial role in weighting our FOM. When calculating the cost function, which in this case is transmission, certain spectral ranges exhibit minimal transmission. Our goal is to improve transmission near the band edge. It is possible that the behavior of the parameters remains unchanged because they are optimized beyond the band edge, effectively acting as a weighting factor in our FOM.

Results

The results in figure 3.33 show the comparison of fixed a and r values between the reference structure, the initial guess, and the optimized structure based on inverse design.

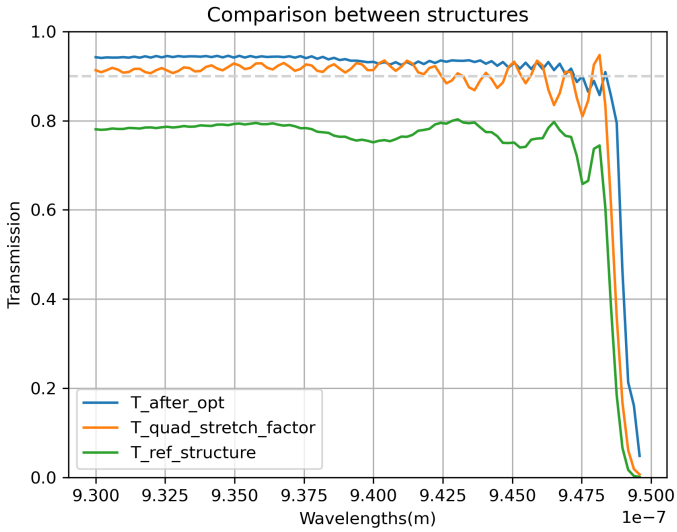


Figure 3.33: Performance Comparison of Different Structures such as Reference structure, Quadratic stretch factor Structure and Optimized Structure

Extrapolation of the behavior

The aim is to know if we can extrapolate the behavior of our optimization for another kind of PhCs with other a and r values. The effect of different a and r values is to change the spectral location of the band edge. A higher hole radius decreases the wavelength of the band edge. For our optimization, we only optimized for $a = 246nm$ and $r = 77nm$. Thus for this extrapolation, we create different geometries and simulate the transmission with a wavelength range between 910 nm and 990 nm in order to obtain the transmission close to the band edge.

We showed that we improved the simulated transmission by using shape optimization and we want to know if it is the case for different a and r values.

The band edge for different a and r values is plotted in figure 3.34.

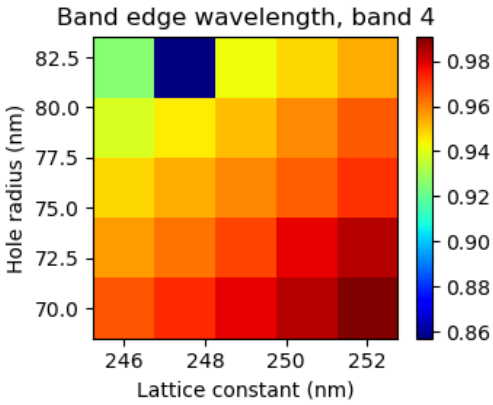


Figure 3.34: Band edge for different a and r values obtained with PWE method

Our optimization is only focusing on a range of wavelengths between 930 nm and 950 nm, it might be possible that the behavior will not be the same for different a and r values due to the band edge.

Thus, we simulated the transmission of both structures. The first one is the reference structure and the other one is the optimized structure with the optimized parameters. Moreover, the ratio's figure will showcase the efficiency of the optimization.

Reference structure / Optimized structure

The reference structure is the current structure that is placed on the SPS chips in use at Sparrow Quantum.

Before plotting the mean transmission for various values of a and r, it is necessary to first fit curves based on a predefined function, as shown in Figure 3.35, and assess the robustness of the fit using the R^2 value. This fit is necessary to calculate the mean transmission of the part which is before the band edge. The function used for fitting the data can be expressed as:

$$T(\lambda) = \begin{cases} 1 & \text{if } \lambda \leq \lambda_{\text{band edge}} \\ 0 & \text{if } \lambda > \lambda_{\text{band egde}} \end{cases}$$

where λ_{band} represents the spectral location of the band edge.

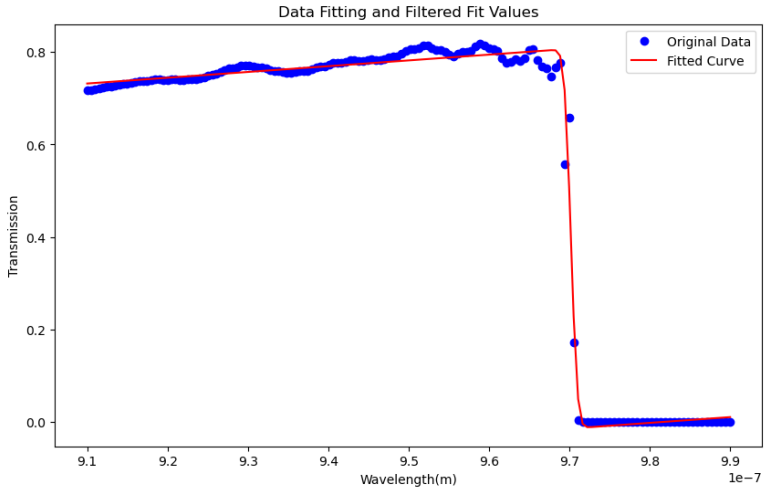


Figure 3.35: Data and curve fitted

Figures 3.36 and 3.37 show an heatmap of R^2 for different parameters of PhC.

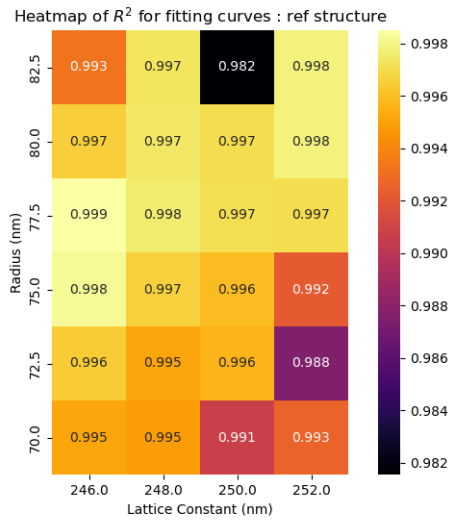


Figure 3.36: Fitting result for reference structure

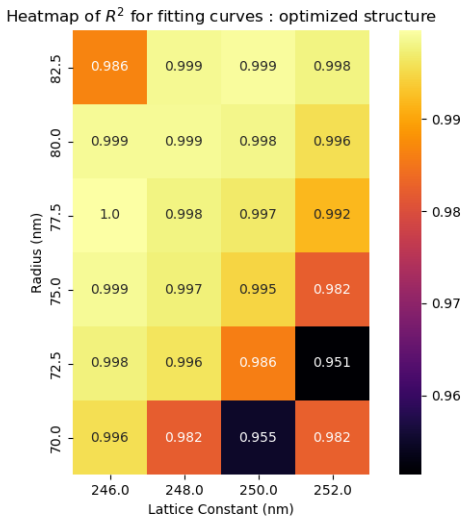


Figure 3.37: Fitting result for optimized structure

We can see that for some fittings the results are not the best but in general as illustrated in figures 3.39 and 3.38. In general, the curve fitting is quite good.

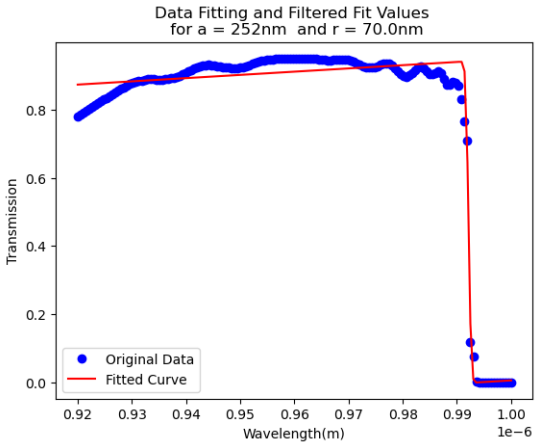


Figure 3.38: Fitting result : worst fit with $R^2 = 0.951$

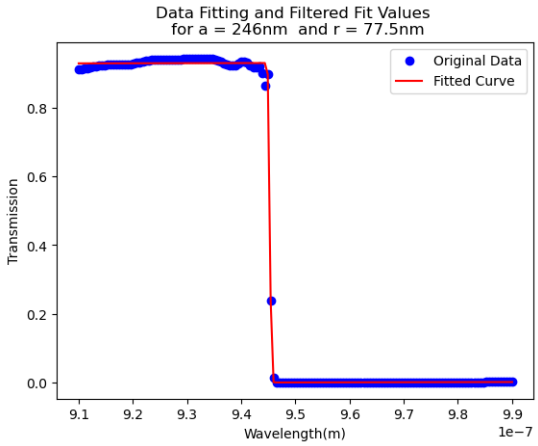


Figure 3.39: Fitting result : best fit with $R^2 = 1.0$

Once, we have fitted the results. We can calculate the mean transmission before the band edge to compare the transmission for different a and r values. Figures 3.40 and 3.41 depicted the mean transmission for reference and optimized structure.

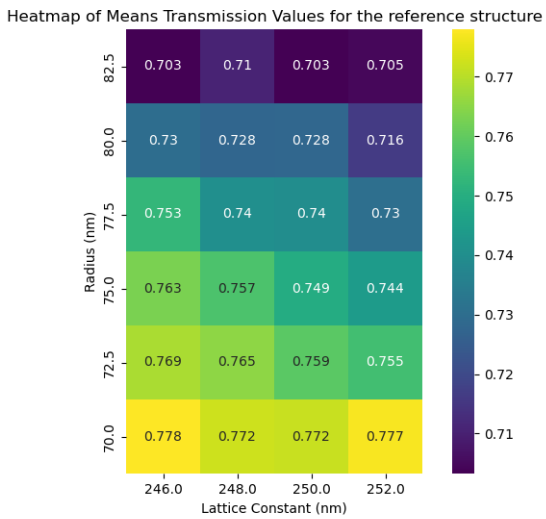


Figure 3.40: Mean Transmission for reference structure

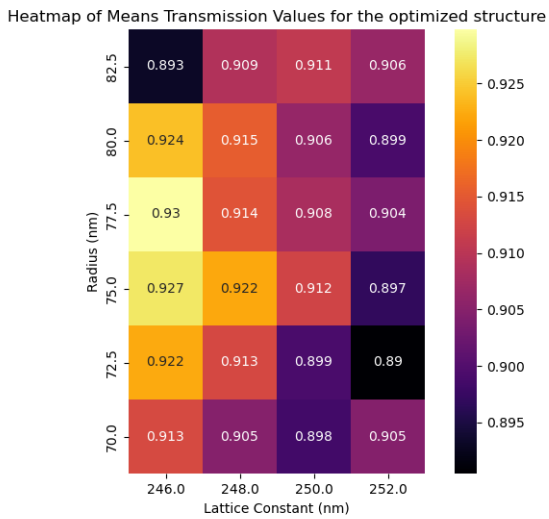


Figure 3.41: Mean Transmission for optimized structure

Optimization ratio's

Since the ratio is larger than 1 in all cases, we conclude that we can extrapolate our optimized parameters onto other a and r values. It can be seen in figure 3.42.

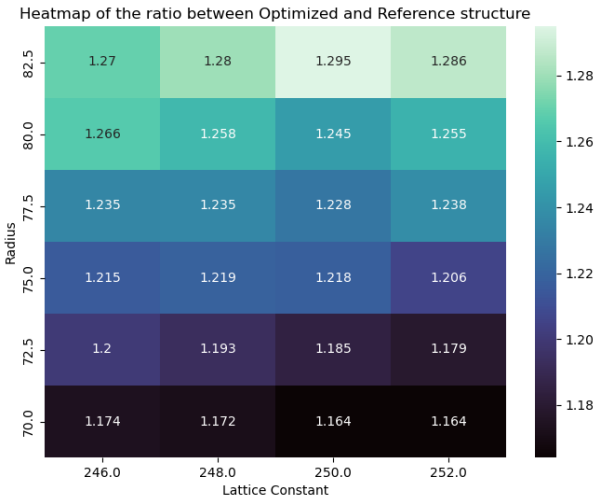


Figure 3.42: Gain of the optimization

Regarding the results for the $a = 252nm$, we want to try another optimization for this setup.

Analysis of PhC Transmission Performance

Before, we mentioned that we do not simulate the device that is used for the emission of a single photon. Indeed, the emission of the QD takes place in a PHC WG in which there is only one interface between the fast section and the nanobeam waveguide. Contrary to this, in our simulation, we have two identical interfaces because our simulation is symmetric. We can note the transmission of this interface: T. It is illustrated in the accompanying figure 3.43.

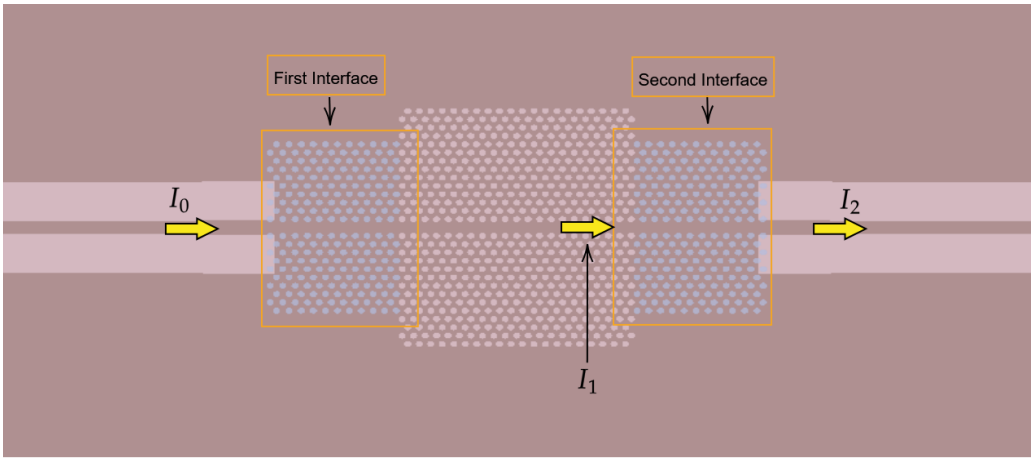


Figure 3.43: Analysis of the simulation

Thus, if we consider a light intensity I_0 at the first interface named T.
Thus, we have $I_1 = T.I_0$ after the first interface and we have $I_2 = T.I_1 = T^2.I_0 = T_{sim}.I_0$ after the second interface.

Regarding, the device used for SPS, we can say that the transmission for the device is

$$T = \sqrt{T_{\text{sim}}} \tag{3.3}$$

because there is only one interface in the component on the chip and the two interfaces that we simulate are truly the same. That's why, we can take the square root of T_{sim}. The plots before show T_{sim} and finally we want to know the transmission T of the PhC WG. The gain of transmission in percentage is displayed in figure 3.44.

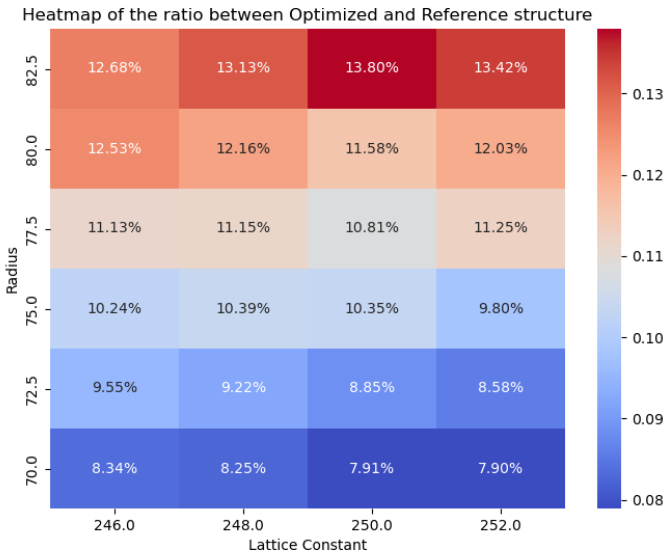


Figure 3.44: Gain of transmission which contributes to η_{Chip}, in percentage

CHAPTER 4

DISCUSSION

4.1 Improvement of the current optimization

During my internship, the optimization was done for fixed a and r values of PhCs. The values are $a = 246nm$ and $r = 77nm$. It could be interesting to try another optimization for other values. Indeed, in our optimization, we focused on a range of wavelengths which is not the same for different a and r values regarding the band edge. Maybe, it will interesting to try an optimization for different lattice values.
As we said, in 3.1, we want to avoid an abrupt interface with these two sections of the PhC WG and make an adiabatic transition. However, in our optimization, the optimizer gave us a big stretch factor at the beginning. It could be interesting to change the boundary condition for the first parameter.

An improvement could be to try an optimization in which you narrow the range of wavelengths in order to improve the transmission in a very specific part of the wavelengths.

Another improvement could be to try a topology optimization in which you place the solver region between the NW and the fast region. We need to be very careful about the conditions on the solver. The manner in which you apply conditions to the solving region can influence the optimization results.

4.2 Using for other parts of the SPS

Currently, inverse design, and more specifically topology optimization, is frequently employed for designing waveguides to minimize losses. If certain areas of the device require enhancement, it may be worthwhile to explore this optimization method, as the existing framework is now fully equipped and ready for implementation. One aspect of the component that could be optimized is the shallow-etched grating outcoupler[21] where you collect the photon, as its current efficiency

leaves room for improvement. This efficiency is crucial for the overall single-photon source efficiency η_{SPS}, as it directly affects the coupling efficiency η_{Coupling}.

Conclusion

During this internship, I had the fascinating opportunity to delve into the world of research in a growing start-up affiliated with a well-known institute of Research in a field that I like very much. My primary focus was on a specific part of the PhC development project, I had the chance to engage in diverse tasks, allowing me to acquire valuable skills in various fields, including data analysis, coding with Python, simulating with Lumerical, and observing some lab experiments.

Throughout the experience, I quickly realized the significance of thorough documentation reading and the time it takes to grasp complex theoretical concepts. The experience of speaking English every day during the internship was very helpful for me to improve my English speaking and comprehension.

Yet I found great value in the regular team meetings, which provided invaluable guidance, feedback, and a deeper understanding of the all project. Despite the brevity of the internship, I was elated to witness some results of my own contributions, making the journey all the more rewarding.

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APPENDIX A

CODE OPTIMIZATION

Based on lumerical example, 2D grating coupler from Ansys Website.

```
import numpy as np
from lumapi import FDTD
from lumopt.geometries.parameterized_geometry import ParameterizedGeometry
from lumopt.utilities.wavelengths import Wavelengths
from lumopt.figures_of_merit.modematch import ModeMatch
from lumopt.optimizers.generic_optimizers import ScipyOptimizers
from lumopt.optimization import Optimization


def run_phc_optimization(Phc, initial_params: list, bounds: list):
    def gen_phc(params: list, fdtd: FDTD, only_update: bool):
        if not only_update:
            Phc.setup_geometry(params, fdtd)
        else:
            Phc.update_geometry(params, fdtd)

    geometry = ParameterizedGeometry(func=gen_phc,
                                     initial_params=initial_params,
                                     bounds=bounds,
                                     dx=1e-5,
                                     deps_num_threads=1)

    wavelengths = Wavelengths(0.93e-6, 0.95e-6, 100)

    fom = ModeMatch(monitor_name="fom",
                    mode_number="Fundamental TE mode",
                    direction="Forward",
                    target_T_fwd=lambda wl: np.ones(wl.size),
                    norm_p=1,
                    target_fom=1.)

```

```
    optimizer = ScipyOptimizers(max_iter=250,
                                method='L-BFGS-B',
                                scaling_factor=1,
                                ftol=1e-6,
                                pgtol=1e-6)


    opt = Optimization(base_script=Phc.setup_script,
                       wavelengths=wavelengths,
                       fom=fom,
                       geometry=geometry,
                       optimizer=optimizer,
                       hide_fdtd_cad=False,
                       use_deps=True)

    return opt.run(working_dir="./optim_export")
```

APPENDIX B

INVESTIGATION OF FIELD PENETRATION REGARDING THE SECOND MODE

There is another way of optimizing the structure which is mentioned in [15] and it improves a lot the transmission. It has been already tested by the company. This try is named a "limited bowtie" shape. However, there is a potentially odd mode that can overlap with the fundamental mode and it can affect the field environment of the QD. Indeed, the penetration of the mode 2 in fast region is higher in this kind of structure compared to the 2 others.

Here we have the plot of the mean transmission which shows the improvement of transmission

Figure B.1: Limited Bowtie Shape

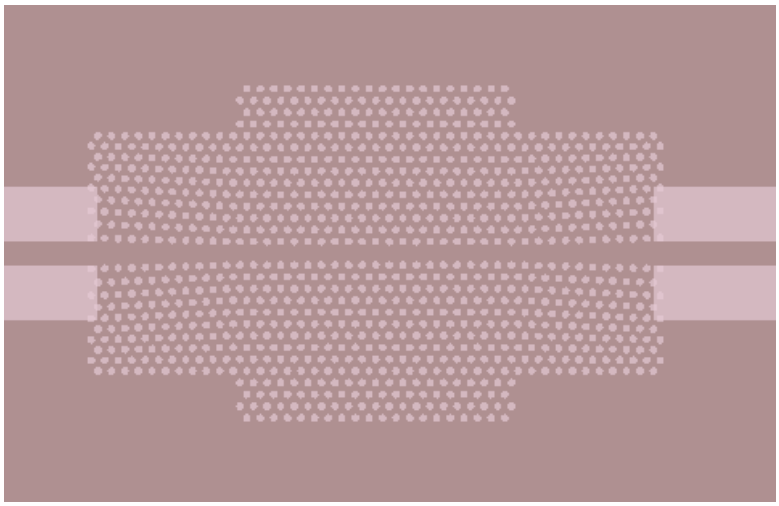
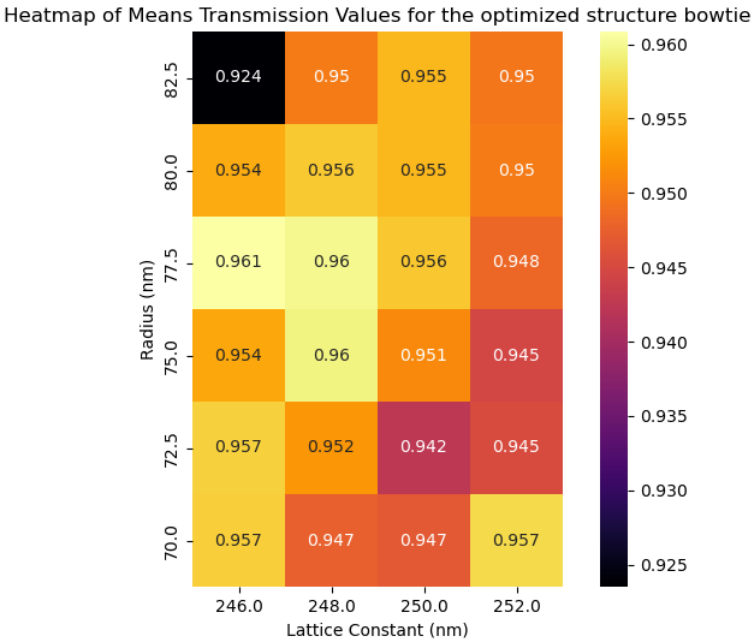


Figure B.2: Mean Transmission Bowtie Shape



APPENDIX C

LIBRARIES USED FOR DATA ANALYSIS

The following Python libraries were used for data analysis in this project:

- **pandas**: Used for data manipulation and analysis, especially for working with structured data.
- **glob**: Utilized to retrieve files/pathnames matching a specified pattern.
- **matplotlib.pyplot**: Employed for creating static, animated, and interactive visualizations.
- **seaborn**: An interface for drawing heatmaps.
- **re**: The regular expression library used for string matching and manipulation.
- **numpy**: Fundamental package for numerical computing with support for arrays and matrices.
- **lmfit.models**: Used for curve fitting with models like `LinearModel` and `StepModel`.
- **sklearn.metrics**: Utilized for evaluating models, specifically the `r2_score` function to assess the goodness-of-fit.